ALGEBRA 1 - UNIT 2

Section 2.1
Rate of Change and Slope

Measuring how one quantity changes compared to another.

> **Mathematical idea:** Slope is not just a number. It tells how fast a relationship changes and what that change means.

Today we will connect slope to graphs, tables, ordered pairs, and real situations. We will focus on the meaning of the change, not just the calculation.

Learning Target

Students will interpret slope as a measure of constant rate of change.

I can statements

- I can calculate slope from graphs.
- I can calculate slope from tables.
- I can interpret slope in context.
- I can determine whether a relationship has a constant rate of change.
- I can connect slope to unit rate and change.

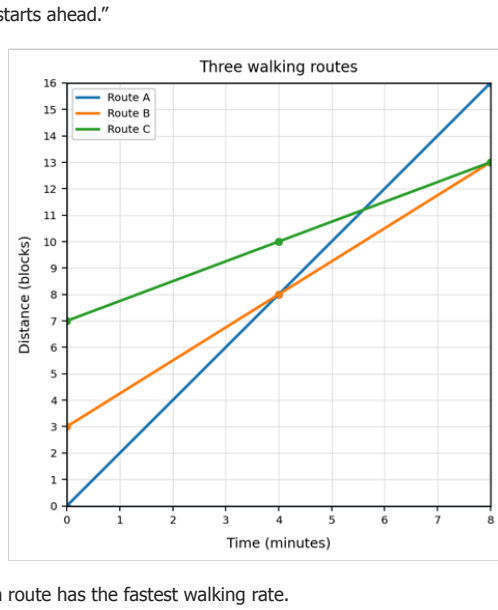
What's To Come

This is a long-range preview for future lessons. Do not solve it today. Instead, annotate it individually. Circle anything familiar, underline symbols or directions that seem important, and write notes about what information you think will be needed later.

As you annotate, ask yourself: What is changing? Which route has the greatest rate of change? Does starting ahead always mean finishing ahead? What evidence from a graph, table, equation, or context could support a recommendation?

Future Task

Three students are comparing walking routes from school to a park. The graph shows distance traveled over time. A student says, "Route C is best because it starts ahead."



1. Use evidence to decide which route has the fastest walking rate.
2. Explain why a route can start ahead but not have the greatest rate of change.
3. Write a recommendation that depends on the amount of time someone plans to walk.
4. Explain what information would make a graph-based recommendation stronger.

Intro

Directions

Read the introduction aloud with a partner or in a small group. As you read, annotate the text by highlighting important ideas, circling key vocabulary, and writing short notes about connections, questions, or predictions in the margins.

A **rate of change** compares how one quantity changes when another quantity changes. If a person earns the same amount of money each hour, the rate of change tells how many dollars they earn for each hour.

In a graph, we often call this rate of change the **slope**. Slope compares the vertical change to the horizontal change. You may hear this described as "rise over run." The rise is the change in output. The run is the change in input.

A **constant rate of change** means the relationship changes by the same amount again and again. In a table, equal input changes should create equal output changes. On a graph, a constant rate of change appears as a straight line.

Slope is most useful when we can explain it in context. A slope of 4 might mean 4 dollars per hour, 4 miles per gallon, or 4 gallons per minute. The number matters, but the units and meaning matter too.

Stop and Discuss

1. What does a rate of change compare?
2. How are "rise" and "run" connected to input and output?
3. Why is it not enough to say only "the slope is 4" in a real situation?

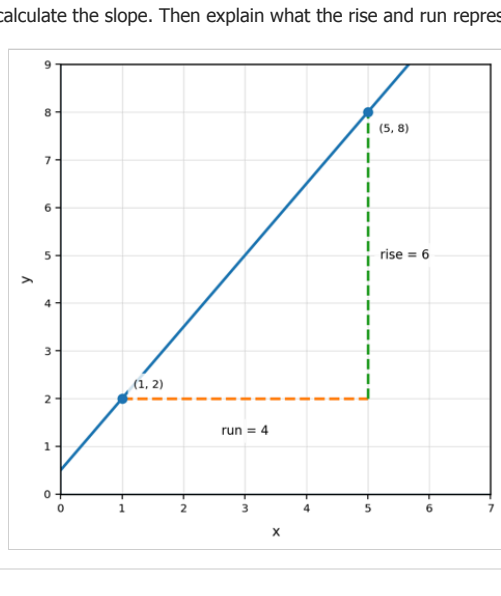
Slope Measures Vertical Change Compared to Horizontal Change

On a graph, slope compares the change in output to the change in input. A slope triangle can help you see both changes.

$$\text{slope} = \frac{\text{change in output}}{\text{change in input}} = \frac{\text{rise}}{\text{run}}$$

Example 1

Use the graph and slope triangle to calculate the slope. Then explain what the rise and run represent.



You Try It 1

A line passes through the points $(2, 3)$ and $(6, 11)$. Find the change in output, the change in input, and the slope. Explain what each part of your calculation means.

Stop and Discuss

1. Why does the order of the subtraction matter when finding change?
2. How can a slope triangle help you avoid mixing up rise and run?

Tables Show Whether the Change Is Constant

A table can show slope by comparing how the output changes as the input changes. When the input changes by equal steps, look for equal changes in the output.

Example 2

Find the rate of change from the table. Decide whether the rate of change is constant.

Time (hours)	0	2	4	6
Distance (miles)	5	17	29	41

You Try It 2

A water tank is being filled. Use the table to find the rate of change and decide whether it is constant.

Time (minutes)	0	3	6	9
Water (gallons)	8	20	32	44

Slope Has Units and Meaning

In a context, slope is a unit rate. It tells how much the output changes for each 1-unit change in the input.

Interpretation frame: For every 1 unit increase in the input, the output changes by the slope amount.

Example 3

A babysitter earns the same amount each hour. The table shows total earnings.

Hours	1	2	3	4
Earnings	$14	$28	$42	$56

1. Find the slope.
2. Interpret the slope in context.
3. Explain why the units matter.

You Try It 3

A cyclist travels at a constant speed. After 2 hours, the cyclist has traveled 26 miles. After 5 hours, the cyclist has traveled 65 miles.

1. Find the rate of change.
2. Interpret the rate of change in context.
3. Write one sentence explaining why this rate is a unit rate.

Stop and Discuss

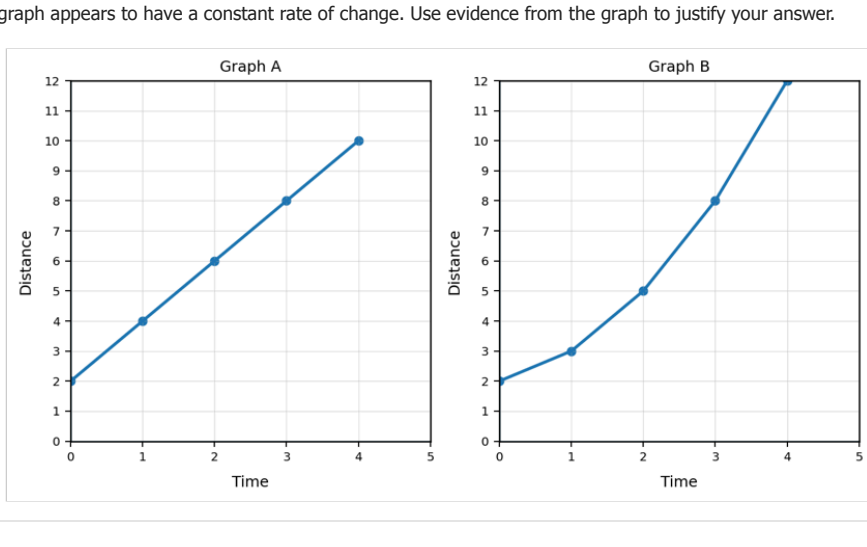
1. What units should the babysitter's slope have?
2. How would the interpretation change if the input were minutes instead of hours?

Constant Rate of Change Means the Pattern Stays Steady

Not every relationship has a constant rate of change. When the change keeps shifting, one slope cannot describe the whole relationship.

Example 4

Decide which graph appears to have a constant rate of change. Use evidence from the graph to justify your answer.



You Try It 4

A student says this table has a constant rate of change because the input increases by 1 each time. Critique the statement.

Input	0	1	2	3	4
Output	2	5	9	14	20

Stop and Discuss

1. Why do equal input changes matter when checking for constant rate of change?
2. Why must you also check the output changes?

Summary

- Rate of change compares how one quantity changes compared to another.
- Slope is the rate of change of a line.
- On a graph, slope can be read as rise over run.
- In a table, a constant rate of change shows equal output changes for equal input changes.
- In context, slope should be interpreted with units.
- A straight line shows a constant rate of change.
- A relationship is not linear if the rate of change does not stay constant.

Common Mistakes

- **Using only rise:** Slope compares rise to run, not just how far the graph goes up.
- **Mixing up input and output:** Change in output goes on top; change in input goes on bottom.
- **Ignoring units:** A slope in context needs meaning, such as miles per hour or dollars per ticket.
- **Assuming every increasing pattern has constant rate:** The output must change by the same amount for equal input changes.
- **Subtracting in different directions:** If you subtract one coordinate first for the output, use the same order for the input.
- **Thinking starting value is slope:** Starting value tells where the relationship begins; slope tells how it changes.

Optional Future Task Reflection

Optional Reflection: Look back at the future task. Do not solve it yet.

1. Highlight one part that makes more sense now than it did at the beginning.
2. Circle one part that still feels unfamiliar.
3. Write one sentence about what representation you would look at first.